CLOUD COMPUTING LAB
LAB REPORT – EXPERIMENT 7

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Section	B
Course Code	CSE12045
Course	Cloud Computing Lab
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Experiment No.	7
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Experiment 7: Familiarization and Usage of Collaborative Applications (SaaS)

1. Aim

To become familiar with Software as a Service (SaaS) and to perform basic collaborative activities using cloud-based productivity and communication applications.

2. Objectives

- To understand the concept and characteristics of the SaaS cloud service model.
- To access a cloud-based collaborative application using an authorised account and identify its features.
- To create, share and collaboratively edit an online document.
- To create, share and collaboratively modify an online spreadsheet and perform basic data operations.
- To create, share and collaboratively edit a cloud-based presentation.
- To explore online communication and collaboration platforms and their features.

3. Requirements

- A computer with an internet-enabled web browser
- An authorised Google account (Google Docs, Google Sheets and Google Slides)
- A collaborator account for sharing and co-editing
- An online communication platform (Google Meet / Microsoft Teams / Zoom)

4. Theory

4.1 What is SaaS?

Software as a Service (SaaS) is a cloud computing service model in which software applications are provided to users over the internet. Users access the application through a web browser or an app without installing and maintaining the complete software infrastructure on their local computers. The

service provider manages the servers, storage, networking, security, application updates and availability.

Examples of SaaS applications include Google Docs, Google Sheets, Microsoft 365, Microsoft Teams, Zoom, Dropbox and Salesforce.

4.2 Common Features of SaaS Applications

- Online access from any location
- Cloud storage of files and data
- Automatic updates managed by the provider
- User authentication and access control
- File sharing with configurable permissions
- Real-time collaboration by multiple users
- Version history and activity tracking
- Access from multiple devices

4.3 Position of SaaS in the Cloud Service Models

Model	What the user manages	Example
IaaS	Operating system, middleware, applications	Amazon EC2
PaaS	Application and data	Google App Engine
SaaS	Only the use of the application and its content	Google Docs, MS Teams

5. Procedure and Observations

Task 1: Introduction to SaaS and Account Setup

Step 1: Access a Cloud-Based SaaS Application

1. Opened a web browser.
2. Accessed the cloud-based SaaS platform (Google Docs).
3. Signed in using an authorised Google account.
4. Explored the application homepage.
5. Identified the options available for creating, opening and sharing files.

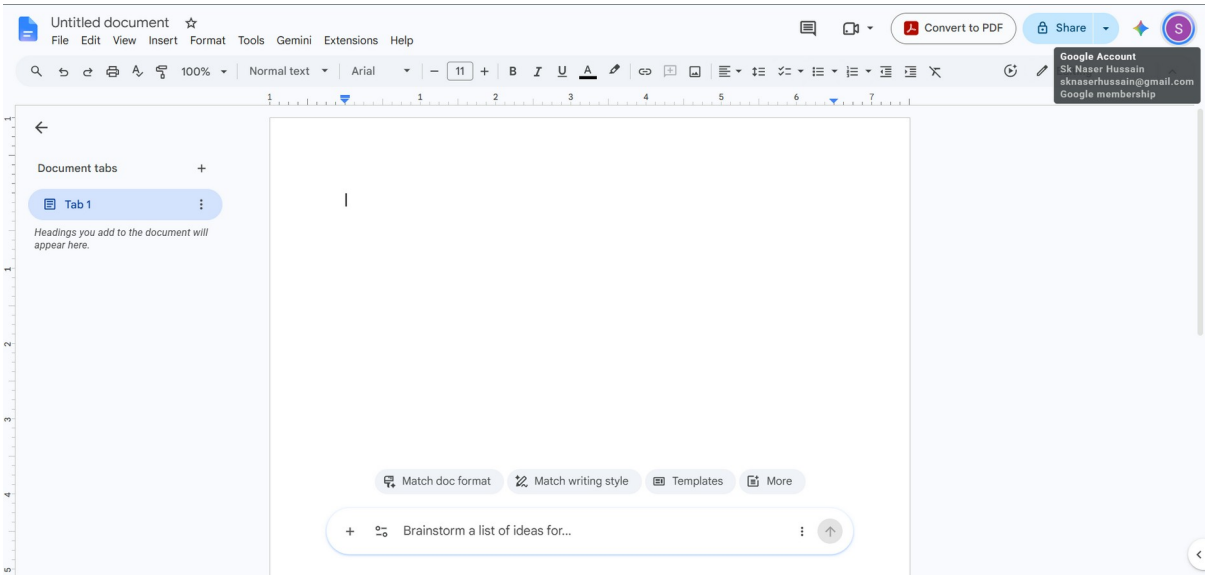


Figure 1: Successful login to Google Docs – signed-in account shown in the profile card

Step 2: Explore SaaS Features

S.No.	Feature	Observation
1	User account / profile options	The Google Account card shows the signed-in user's name and email address (Figure 1).
2	Application dashboard	The editor provides a menu bar, toolbar, document tabs panel and Share button (Figure 1).
3	File creation options	A new blank document is created from the application; the File menu offers New, Open, Make a copy, Download and more (Figure 9).
4	Cloud storage location	Files are stored in Google Drive; the "Saved to Drive" indicator appears beside the title (Figure 13).
5	Sharing options	The Share button opens a dialog to invite people by email (Figure 4).
6	Collaboration options	Multiple users edit simultaneously; comments and live user presence icons are available (Figures 7 and 8).
7	Version / activity history	File → Version history shows saved versions and who edited them (Figures 9 and 10).
8	Settings and access control	Permissions (Viewer, Commenter, Editor) and the Notify people option are set while sharing (Figure 4).

Result: The basic concept of SaaS was understood and the major features of a cloud-based collaborative application were identified.

Task 2: Collaborative Document Creation Using Cloud SaaS

Cloud-based document applications allow multiple users to create, edit and review the same document simultaneously through the internet. Google Docs was used for this task.

Step 1: Create a New Cloud Document

1. Opened Google Docs and signed in with the authorised account.
2. Created a new blank document.
3. Entered the title: Cloud Computing Lab 7 – Collaborative Document.

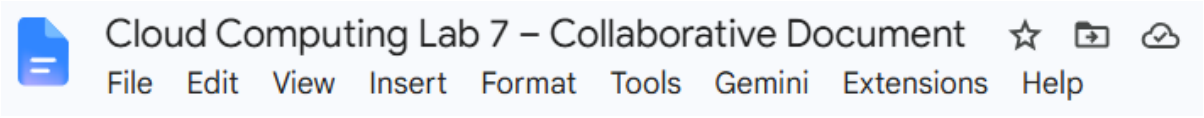


Figure 2: Document titled "Cloud Computing Lab 7 – Collaborative Document"

Step 2: Enter Sample Content

Title: Software as a Service (SaaS)

Content: SaaS is a cloud service model in which software applications are delivered to users through the internet. Users can access the application without managing the underlying infrastructure.

A table describing the SaaS features (Accessibility, Collaboration, Updates and Storage) was also added to the document.

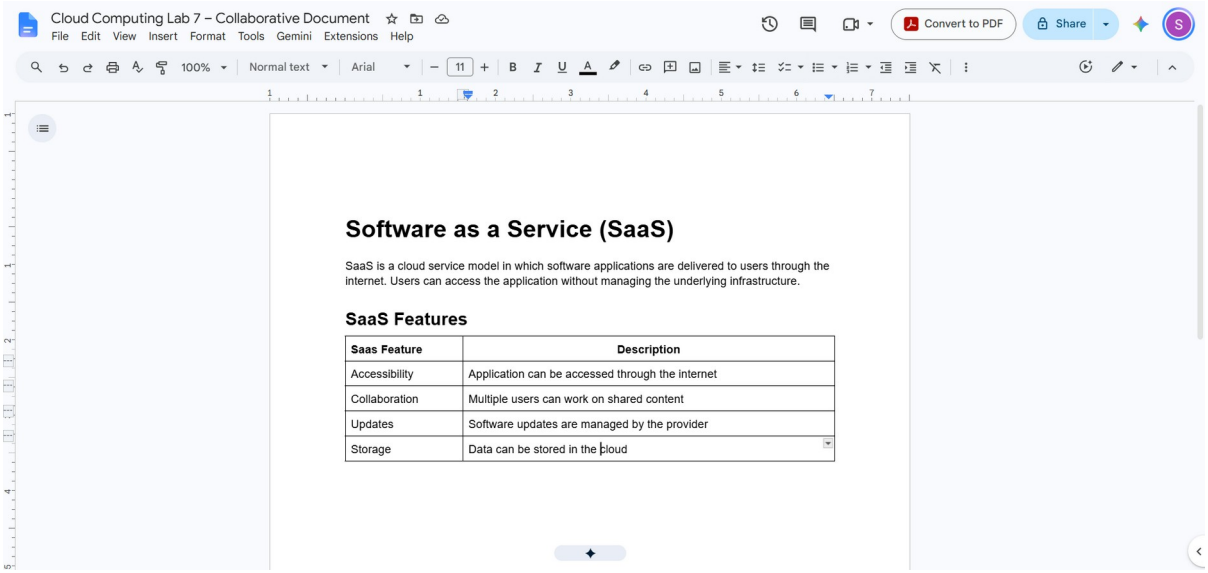


Figure 3: Document with sample content and SaaS feature table

Step 3: Share the Document

1. Selected the Share option.
2. Entered the email address of the collaborator.
3. Selected the Editor permission (Viewer and Commenter are the other available options).
4. Sent the sharing invitation.

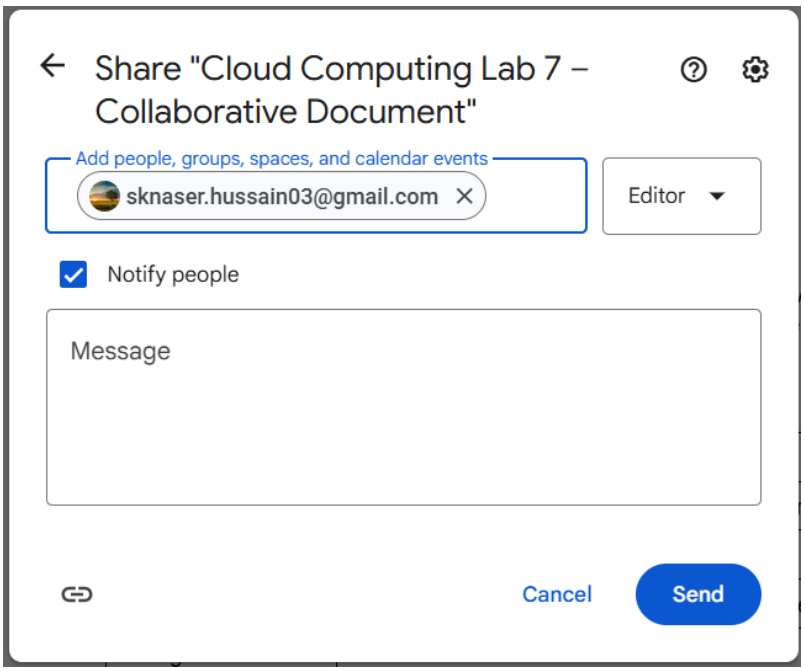


Figure 4: Share dialog – collaborator added with Editor permission

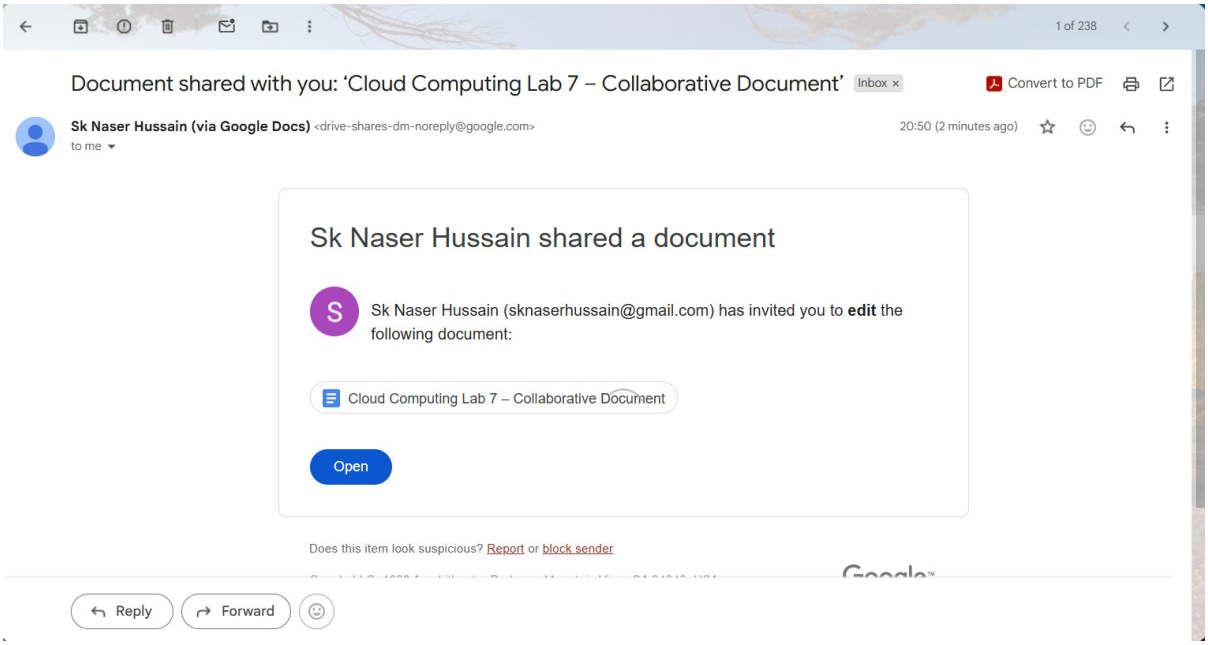


Figure 5: Sharing invitation received in the collaborator's inbox

Step 4: Perform Collaborative Editing

1. The collaborator opened the shared document from the invitation.
2. The collaborator added a new line of text to the document.
3. The collaborator opened the comment box to add a comment.
4. The changes made by the other user were observed in real time.

The collaborator added the line "Collaborative editing was successfully performed using Google Docs." below the table. The edit appeared immediately in the owner's copy, where the collaborator's profile icon and coloured cursor were visible while editing.

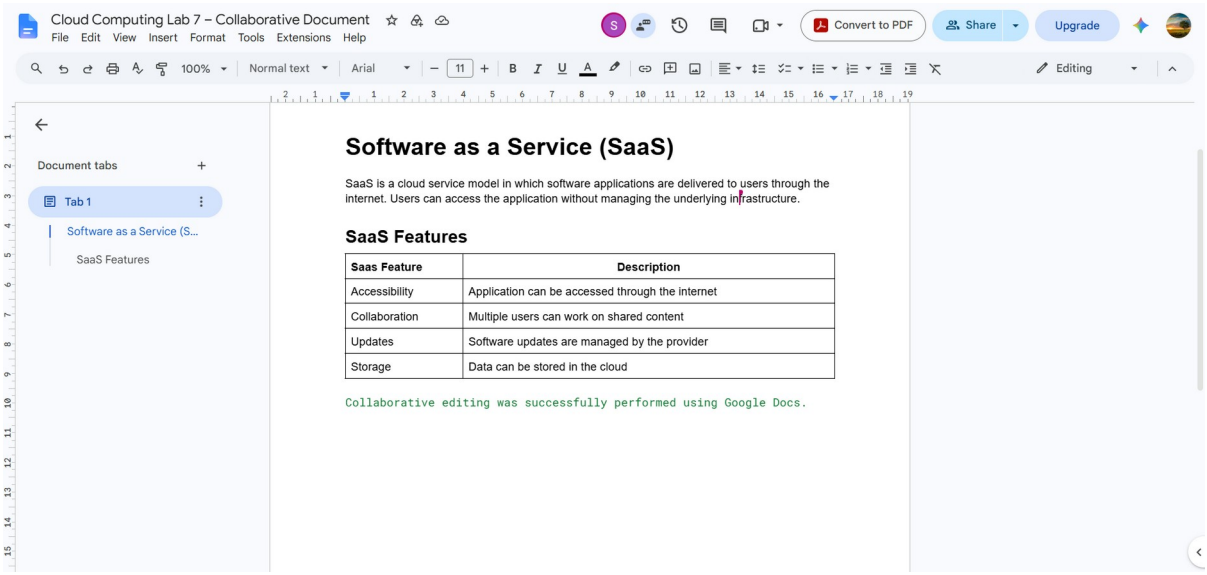


Figure 6: Collaborator's view of the shared document with the new line added

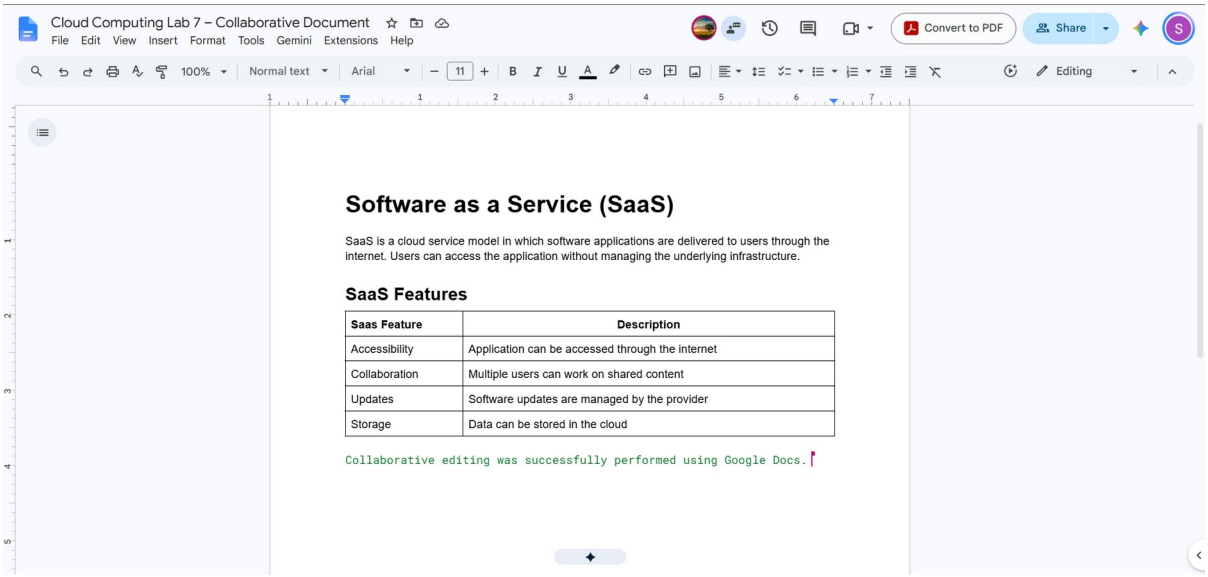


Figure 7: Owner's view showing the collaborator's edit and presence



Figure 8: Adding a comment to the document

Step 5: Observe Version History

1. Opened the document's version history.
2. Observed the changes made by different users.
3. Identified the time and user associated with each change.
4. Observed how previous versions can be identified and restored.

The version history panel listed two saved versions (8:28 PM and 9:00 PM) with the contributing users' names, and highlighted the text added by the collaborator in colour. The panel reported a total of 4 edits, and earlier versions can be opened or restored from it.

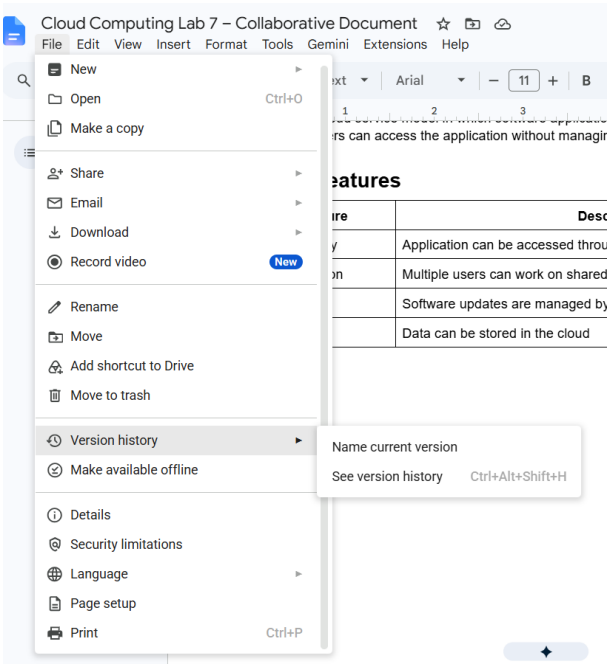


Figure 9: File → Version history → See version history

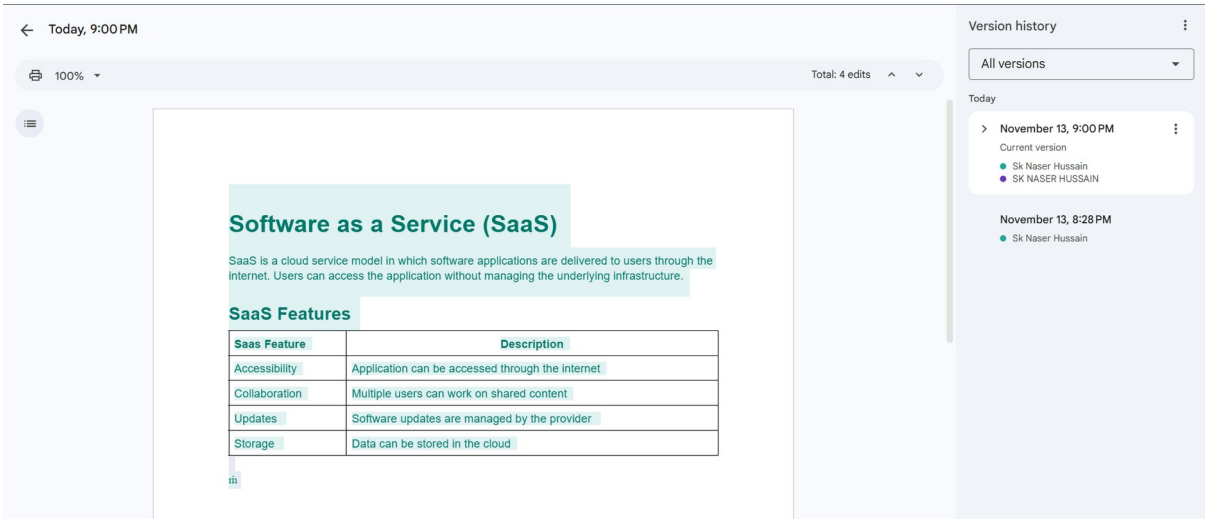


Figure 10: Version history showing edits by different users

Result: A cloud-based document was created and shared with another user. Collaborative editing, commenting and version-history features were demonstrated successfully.

Task 3: Collaborative Spreadsheet Using Cloud SaaS

Cloud spreadsheet applications allow users to store, process and share tabular data through the internet. Google Sheets was used for this task.

Step 1: Create a New Spreadsheet

1. Opened Google Sheets and created a new blank spreadsheet.
2. Entered the title: Cloud Computing Lab 7 – Collaborative Spreadsheet.

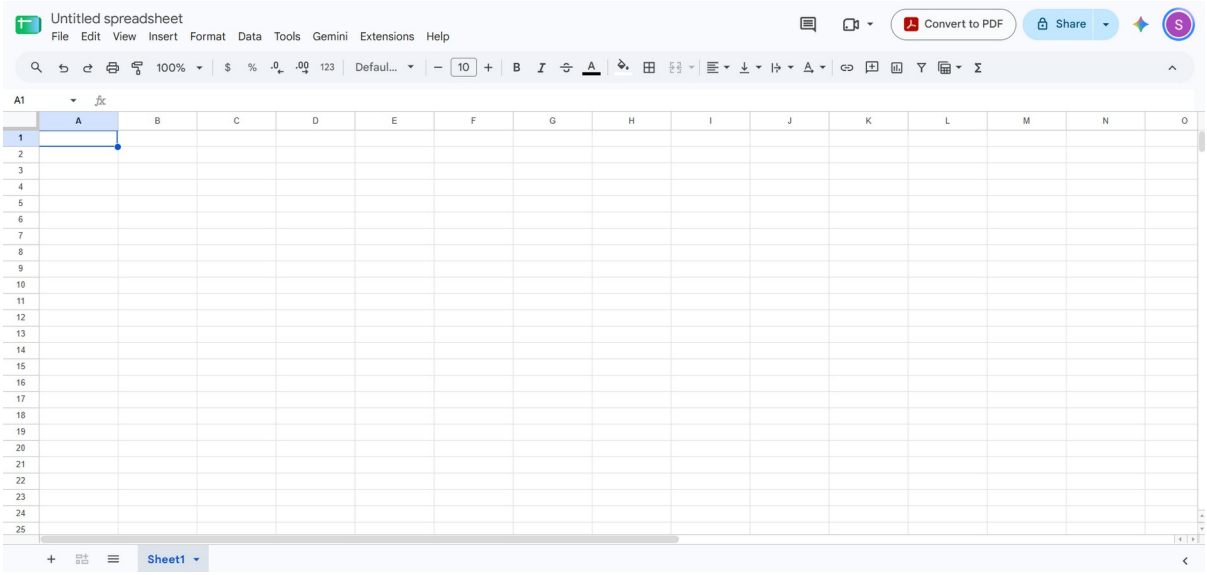


Figure 11: New blank spreadsheet in Google Sheets

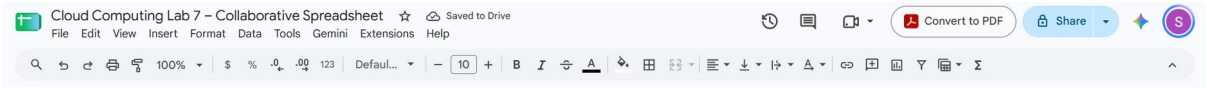


Figure 12: Spreadsheet titled "Cloud Computing Lab 7 – Collaborative Spreadsheet"

Step 2: Enter Sample Data

The following student data was entered (Student ID, Student Name, Subject and Marks) and the header row was formatted in bold with centred marks and borders.

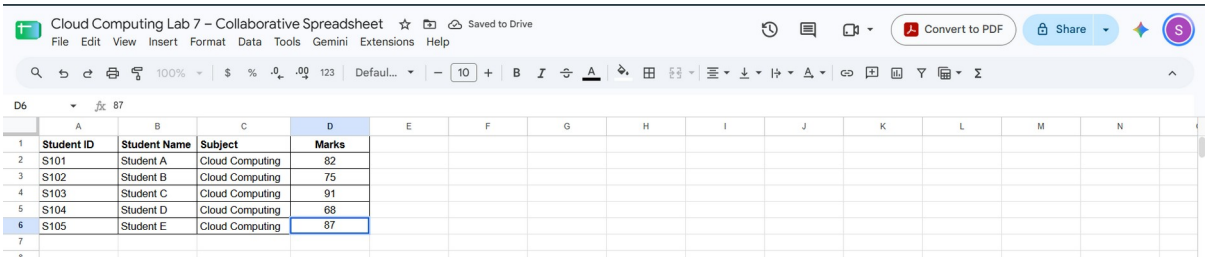


Figure 13: Sample student data entered and formatted

Step 3: Perform Basic Spreadsheet Operations

Operation	Formula	Result
Total marks	=SUM(D2:D6)	403
Average marks	=AVERAGE(D2:D6)	80.6
Highest marks	=MAX(D2:D6)	91
Lowest marks	=MIN(D2:D6)	68

Calculation	403 ×
Total Marks	=SUM(D2:D6)
Average Marks	80.6
Highest Marks	
Lowest Marks	

Figure 14: Total marks using =SUM(D2:D6)

Calculation	Result
Total Marks	80.6 × 403
Average Marks	=AVERAGE(D2:D6)
Highest Marks	
Lowest Marks	

Figure 15: Average marks using =AVERAGE(D2:D6)

Calculation	Result
Total Marks	403
Average Marks	91 × 80.6
Highest Marks	=MAX(D2:D6)
Lowest Marks	

Figure 16: Highest marks using =MAX(D2:D6)

Calculation	Result
Total Marks	403
Average Marks	80.6
Highest Marks	68 × 91
Lowest Marks	=MIN(D2:D6)

Figure 17: Lowest marks using =MIN(D2:D6)

Calculation	Result
Total Marks	403
Average Marks	80.6
Highest Marks	91
Lowest Marks	68

Figure 18: Calculated results table

Step 4: Share the Spreadsheet

1. Selected Share.
2. Added the authorised collaborator.
3. Assigned Editor permission.

4. Shared the spreadsheet.

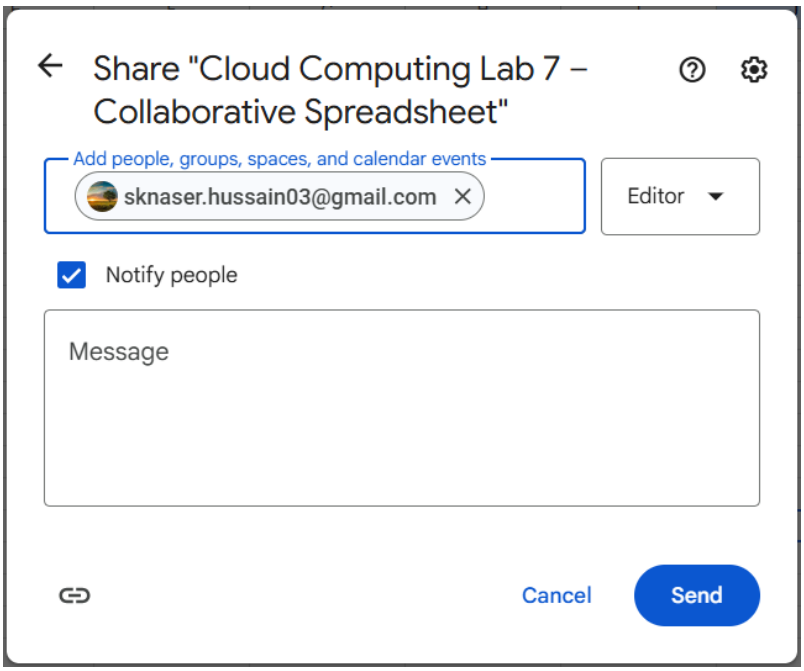


Figure 19: Share dialog – collaborator added with Editor permission

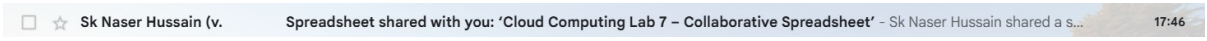


Figure 20: Spreadsheet-shared notification in the collaborator's inbox

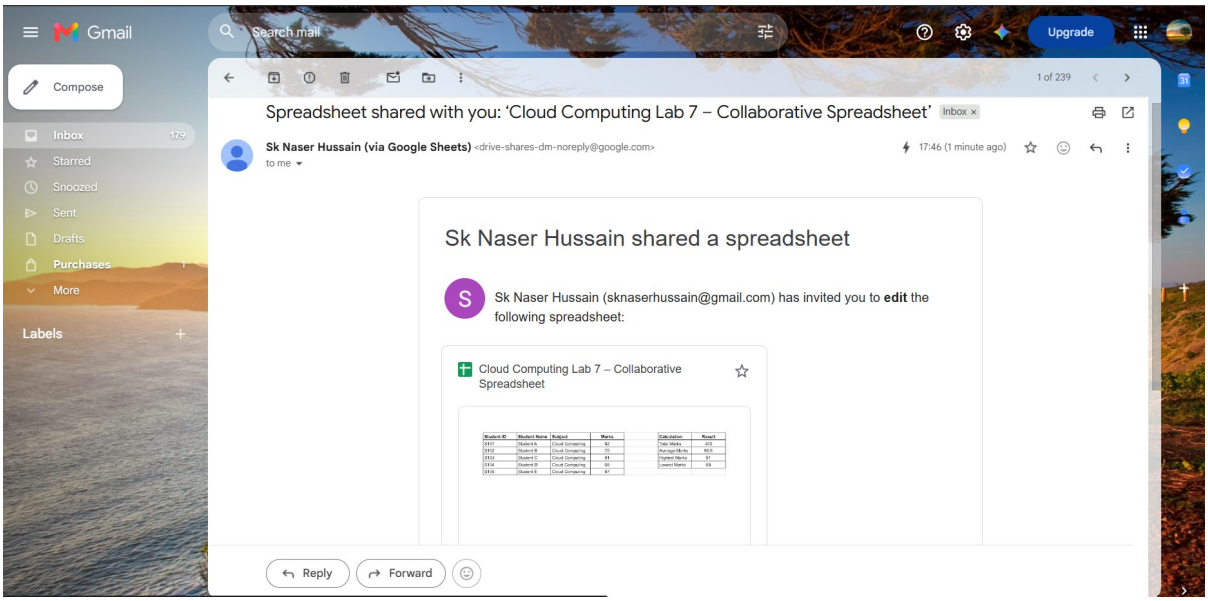


Figure 21: Sharing email opened by the collaborator

Step 5: Perform Collaborative Data Entry

1. The collaborator added another student record (S106, Student F, 85 marks).
2. The collaborator modified one existing value (marks of S101 changed from 82 to 84).
3. The collaborator added a comment to a cell: "New student record added successfully through collaborative editing."
4. The changes made by the other user were observed.

A	B	C	D
Student ID	**Student Name**	**Subject**	**Marks**
S101	Student A	Cloud Computing	84
S102	Student B	Cloud Computing	75
S103	Student C	Cloud Computing	91
S104	Student D	Cloud Computing	68
S105	Student E	Cloud Computing	87
S106	Student F	Cloud Computing	85

Figure 22: New record added and existing value modified by the collaborator

After the collaborative changes, the calculated results updated automatically: total marks 490, average 81.67, highest 91 and lowest 68. The cell comment shows the author's name and the time it was posted.

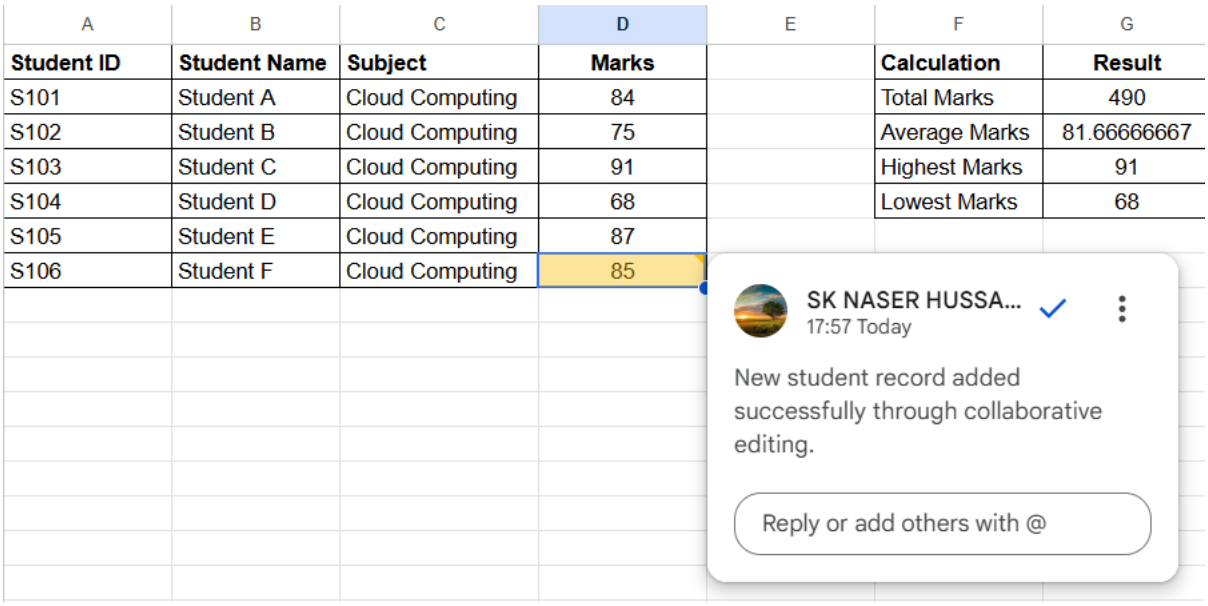


Figure 23: Cell comment and updated results after collaborative editing

Result: A cloud-based spreadsheet was created, shared and collaboratively modified. Basic spreadsheet operations, collaborative data entry and cell comments were demonstrated successfully.

Task 4A: Creating and Collaboratively Editing a Cloud-Based Presentation

Cloud presentation applications allow users to create, edit and present slides through the internet, and multiple users can work on the same presentation. Google Slides was used for this task.

Step 1: Create a New Presentation

1. Opened Google Slides and created a new presentation.
2. Entered the title: Cloud Computing Lab 7 – SaaS Applications.

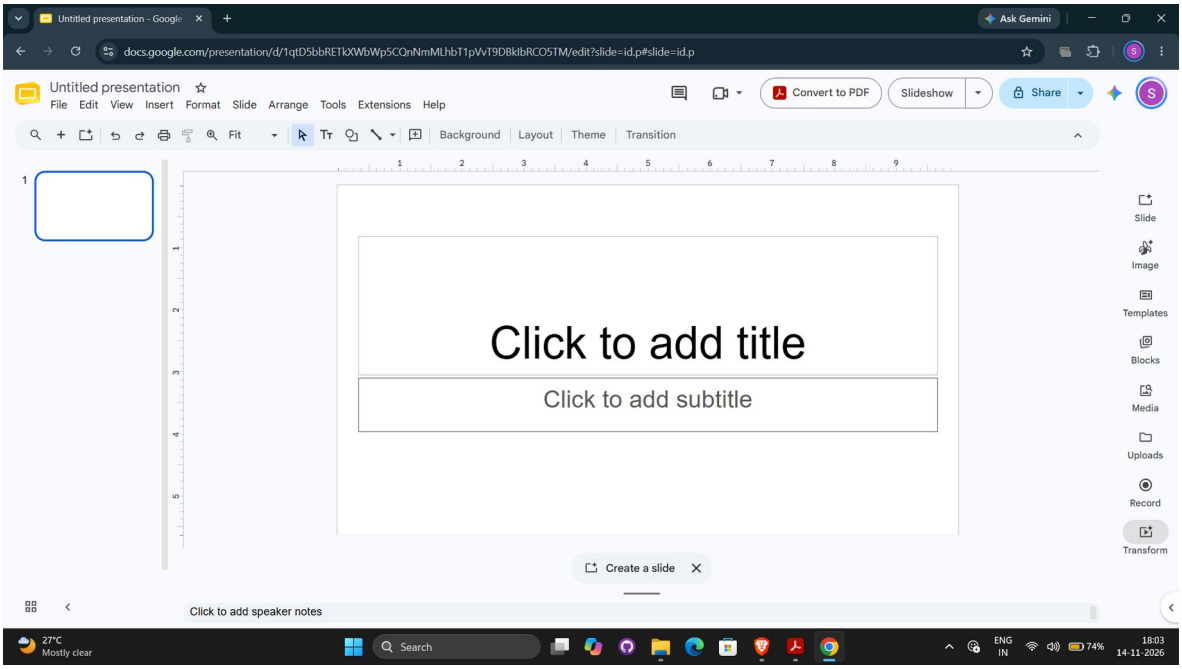


Figure 24: New blank presentation in Google Slides

Step 2: Create Sample Slides

Four slides were created: the title slide, What is SaaS?, Examples of SaaS and Advantages of SaaS.

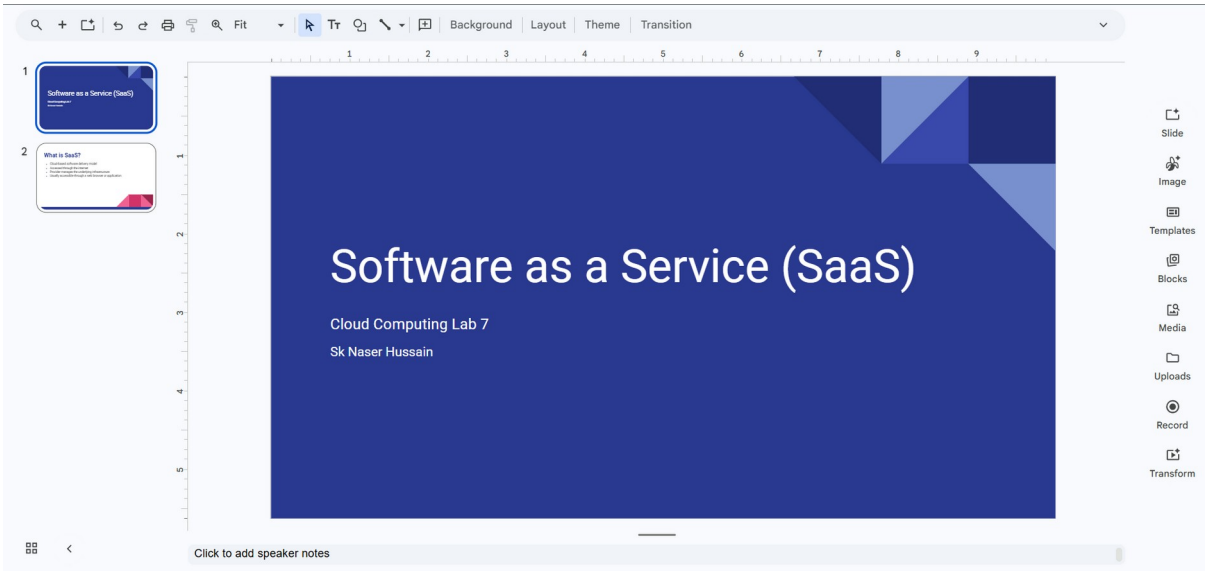


Figure 25: Slide 1: Title slide

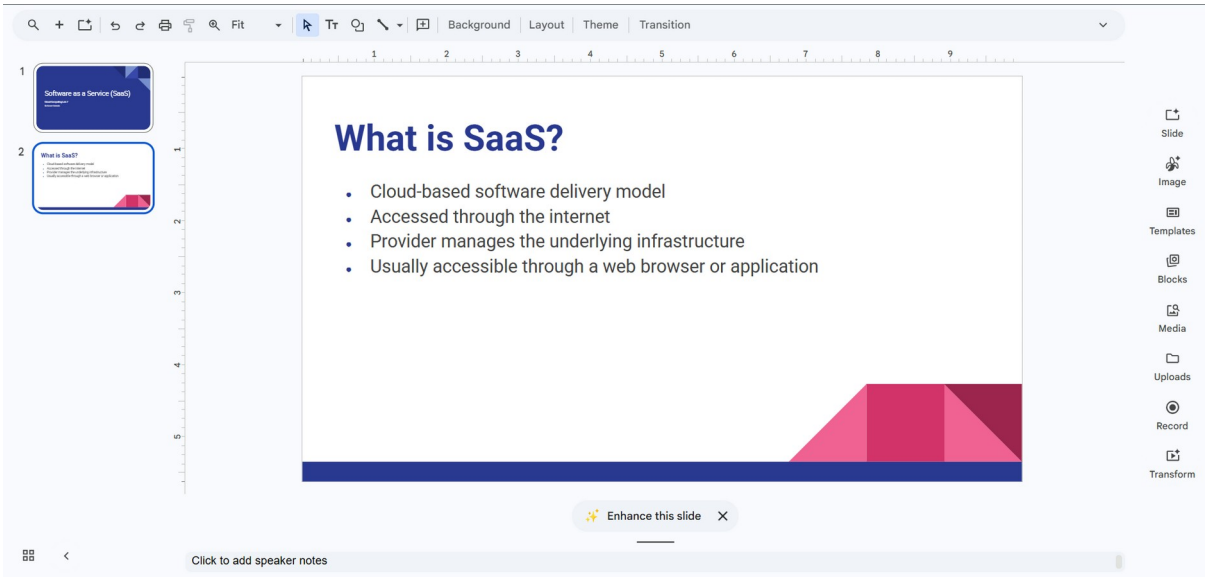


Figure 26: Slide 2: What is SaaS?

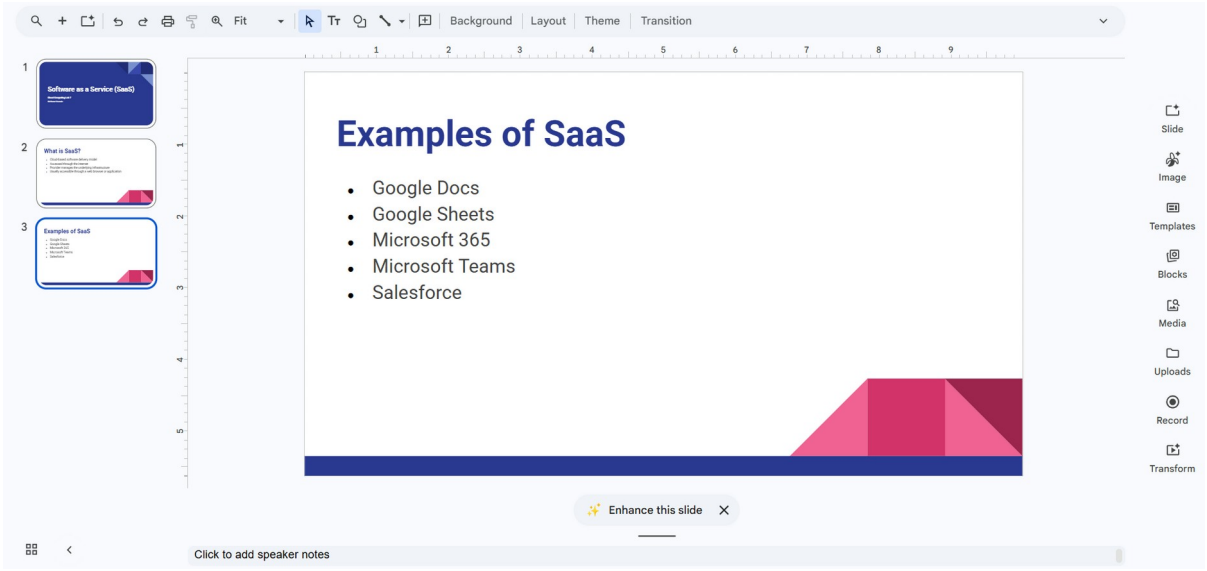


Figure 27: Slide 3: Examples of SaaS

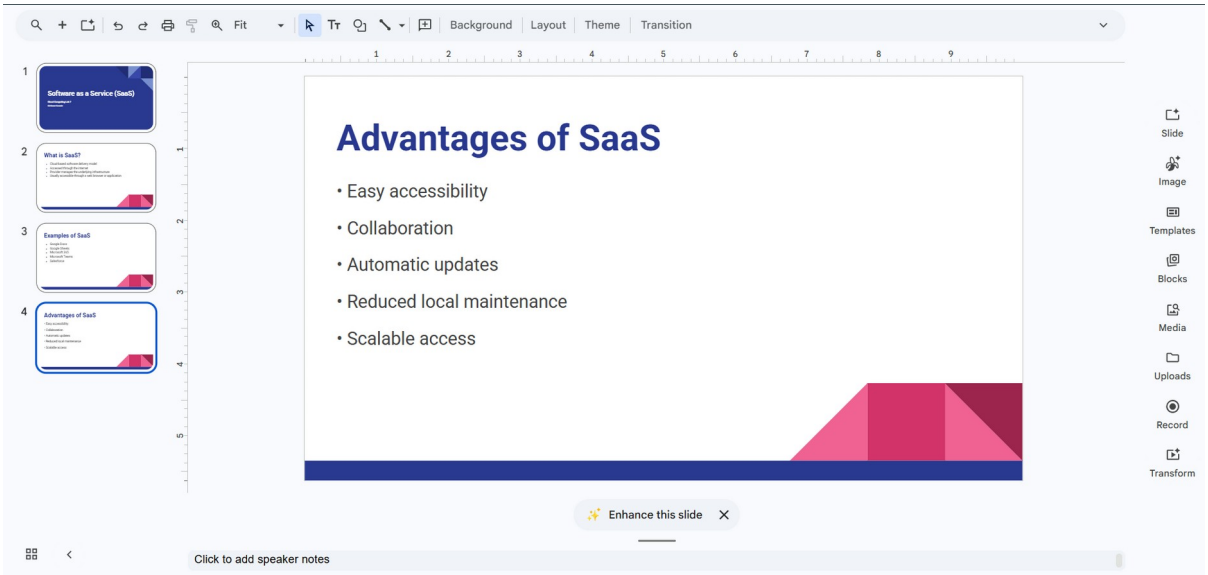


Figure 28: Slide 4: Advantages of SaaS

Step 3: Share the Presentation

1. Selected Share.
2. Added the authorised collaborator.
3. Assigned Editor permission.
4. Shared the presentation.

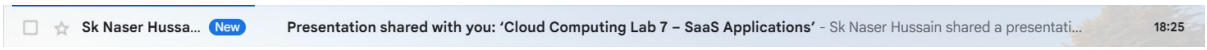


Figure 29: Presentation-shared notification in the collaborator's inbox

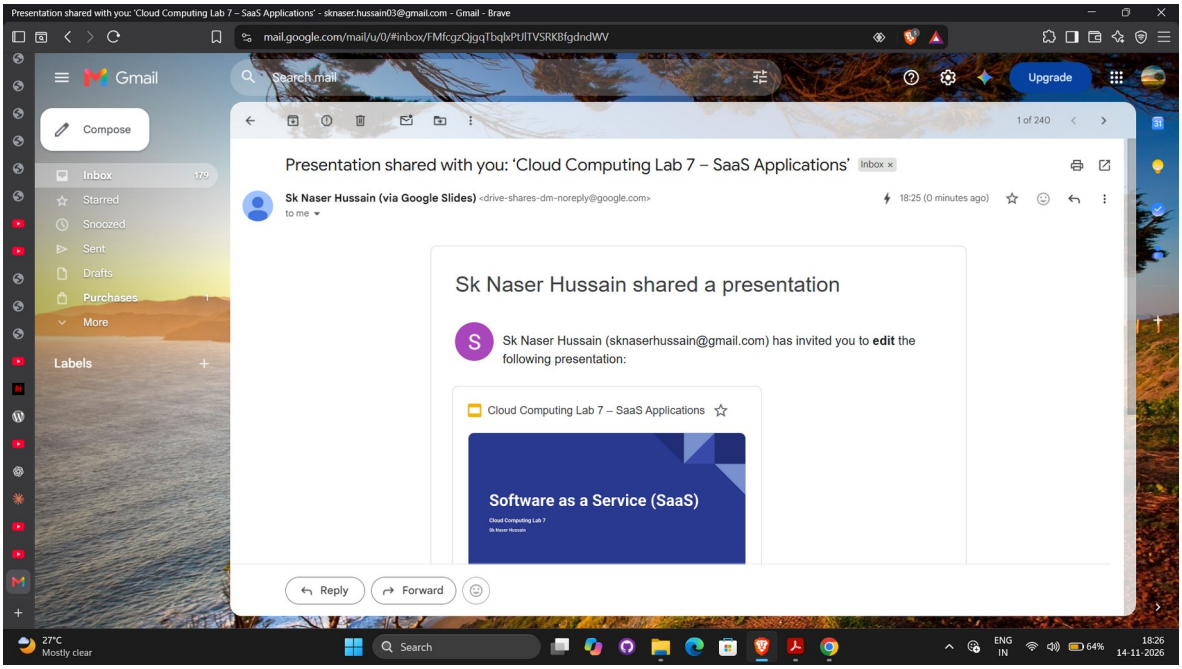


Figure 30: Sharing email opened by the collaborator

Step 4: Collaboratively Modify the Presentation

1. The collaborator opened the shared presentation.
2. The collaborator edited Slide 4 by adding an additional line of text.
3. The collaborator used different formatting (green monospaced text) for the added line.
4. The collaborator added a comment on Slide 3.
5. The changes made by the other user were observed.

The added line "Collaborative editing performed successfully." appeared on Slide 4 with the collaborator's name tag, and the owner's profile and presence icons were visible at the top of the editor.

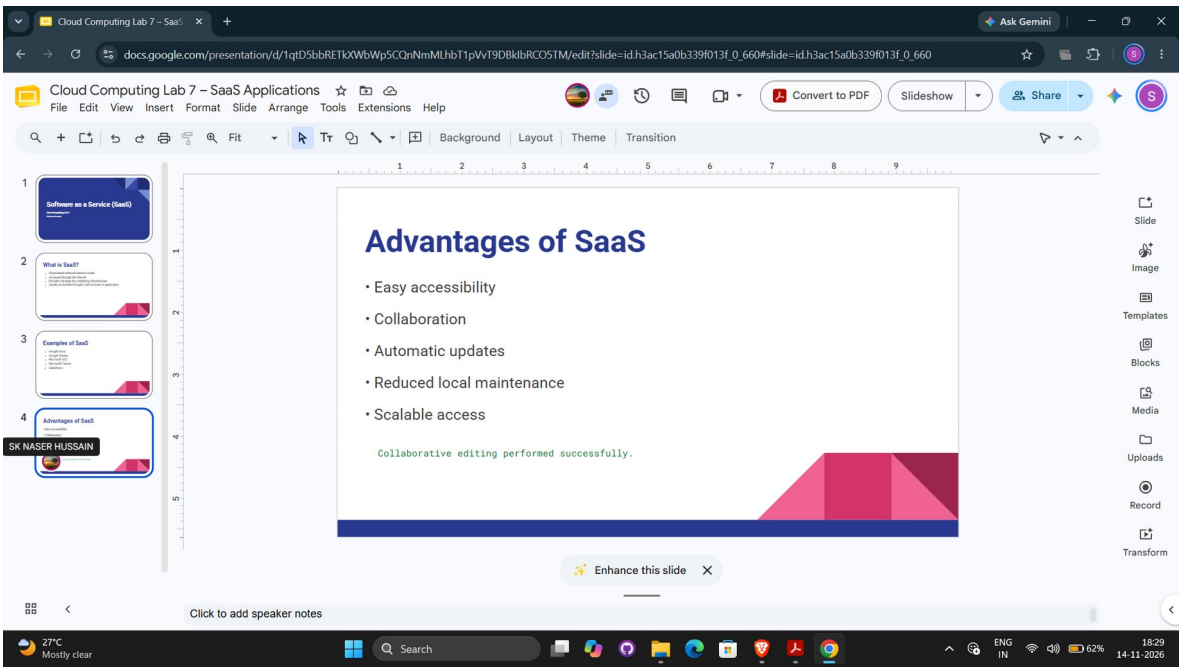


Figure 31: Slide 4 edited by the collaborator

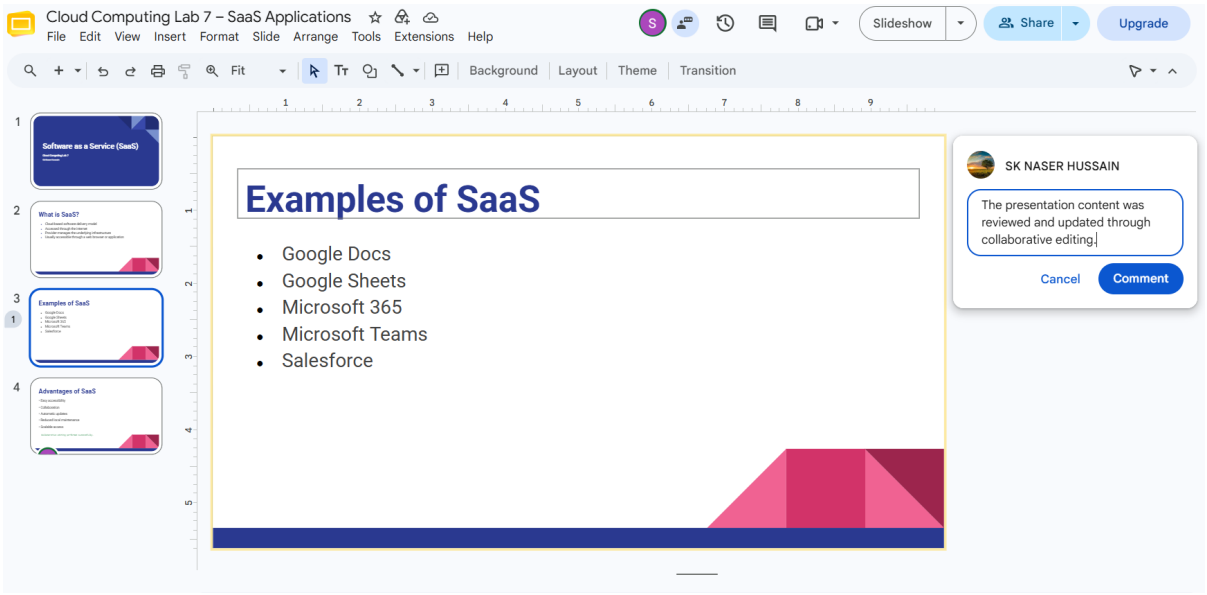


Figure 32: Comment added by the collaborator on Slide 3

Result: A cloud-based presentation was created, shared and collaboratively edited successfully.

Task 4B: Exploring Online Communication and Collaboration

Online communication platforms are SaaS applications that allow users to communicate and collaborate remotely over the internet. Commonly used platforms include Google Meet, Microsoft Teams and Zoom. Like the document, spreadsheet and presentation tools used earlier, they run in a web browser or app, require only a user account, and are fully managed by the service provider.

Step 1: Access an Online Communication Platform

1. Open an authorised online communication platform (Google Meet, Microsoft Teams or Zoom).
2. Sign in using an authorised account – the same Google account used for Docs, Sheets and Slides works with Google Meet.

3. Explore the main dashboard, which provides options to start a new meeting, join a meeting with a code or link, and schedule meetings.

Step 2: Explore Collaboration Features

S.No.	Feature	Description
1	Video and audio meetings	Participants can hold real-time video or voice meetings from any location using a browser or mobile app.
2	Chat and messaging	Text messages can be exchanged with all participants or privately during and after a meeting.
3	Screen sharing	A participant can present a window, tab or the entire screen, so that documents, spreadsheets and slides can be shown live.
4	File sharing	Files and links, such as the shared Google Docs, Sheets and Slides files used in this experiment, can be shared directly in the chat.
5	Participant management	The host can invite, admit, mute or remove participants and control who can present.
6	Recording and captions	Meetings can be recorded and live captions can be enabled, where supported by the platform.
7	Scheduling and calendar integration	Meetings can be scheduled with a link and added to the participants' calendars.

Comparison of Common Communication Platforms

Platform	Provider	Typical use	Key collaboration features
Google Meet	Google	Meetings integrated with Google Workspace	Meeting links, screen sharing, chat, live captions
Microsoft Teams	Microsoft	Team workspace with meetings and chat	Channels, chat, file sharing, Microsoft 365 integration
Zoom	Zoom	Video conferencing and webinars	Breakout rooms, screen sharing, recording, chat

These platforms complement the collaborative documents, spreadsheets and presentations created earlier: a team can discuss the content in a video meeting, present a shared file by screen sharing, and continue editing it in real time with comments, all through SaaS applications and without installing or maintaining any local software.

Result: SaaS-based communication platforms were explored, and their support for remote communication, file sharing and real-time collaboration was understood.

6. Result

An authorised SaaS platform was accessed and its features were identified. An online document, spreadsheet and presentation were created, shared and collaboratively edited, and commenting, real-

time editing and version-history features were demonstrated. Online communication platforms were also explored for remote communication and collaboration.

7. Conclusion

In this experiment, the SaaS service model was studied and cloud-based collaborative applications were used in practice. It was observed that SaaS applications can be accessed through a browser without local installation, that data is stored in the cloud, and that updates are handled by the provider. Features such as real-time co-editing, permission-based sharing, comments and version history make SaaS tools well suited to teamwork.